

DATA SHARDING & ROUTING - QUICK NOTES ☆

① Why split data?

- Distribute data
- Improve availability
- Scale writes & reads
- Reduce hotspots
- Lower latency

Big DB



Split across multiple servers

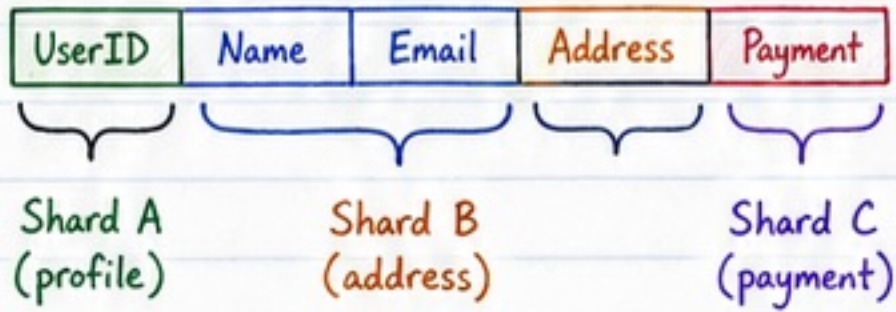


Find data →
Find shard →
Find healthy server

- ✓ Scalable
- ✓ Available
- ✓ Resilient

② Vertical vs Horizontal Partitioning

Vertical = split columns



Horizontal = split rows

UserID	Name	Region
1	A	US
2	B	US
3	C	EU
4	D	EU
5	E	APAC
6	F	APAC

Shard 1
(1-2)

Shard 2
(3-4)

Shard 3
(5-6)

③ What is Sharding?

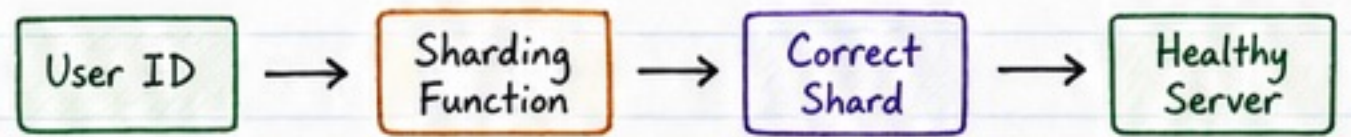
Sharding = rows across independent servers.



- Each shard holds a subset of rows
- Shards live on independent servers
- Adds scale & fault tolerance

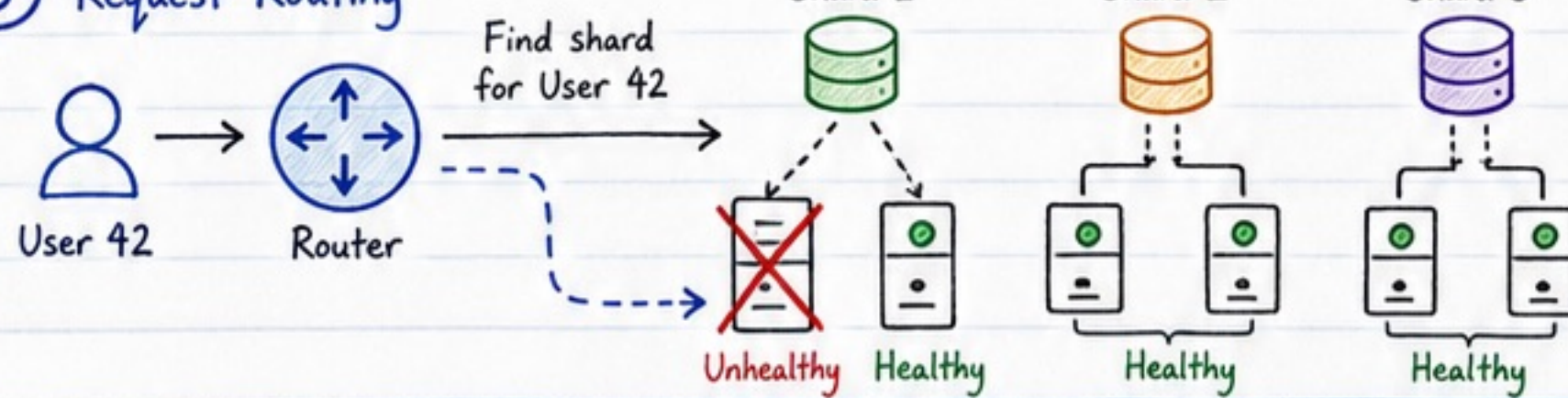
④ Sharding Key

A sharding key decides which shard a row lives on.



User ID → Sharding Function → Correct Shard → Healthy Server

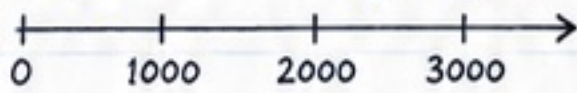
⑤ Request Routing



Route continues to a healthy replica that has the data.

⑥ Routing Strategies

★ Range-based — key ranges



★ Hash-based — $\text{hash}(\text{key}) \% N$

user42 → hash() → % 3 = 1

★ Directory-based — lookup map

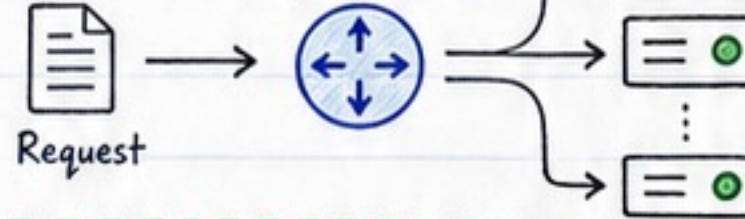
Key Range	Shard
0 - 999	S1
1000 - 1999	S2
2000 - 2999	S3
...	



⑦ Load Balancing vs Sharding

Load Balancing

Which healthy server?



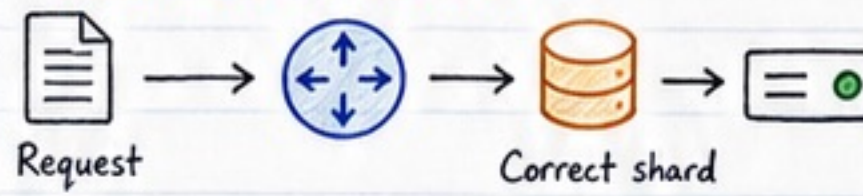
Different questions.
Different goals.

Often work together!



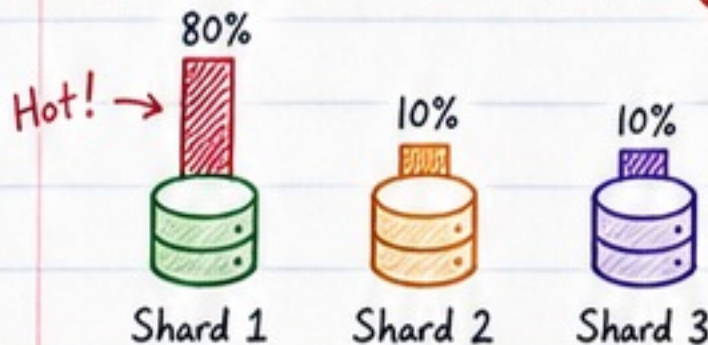
Sharding

Which server has the data?



⑧ Hot Shards + Rebalancing

Hot shard = uneven traffic 🔥



Rebalancing ideas

- ✓ Split hot shard by key range
- ✓ Move ranges to other shards
- ✓ Add new shard
- ✓ Use consistent hashing

Consistent hashing moves fewer keys.

Goals

- ★ Even load
- ★ Low latency
- ★ Smooth movement
- ★ Minimal disruption

GOLDEN RULE

Choose a key with high cardinality and even traffic.

WATCH OUT!

Equal data does not always mean equal load.



INTERVIEW TIP

Keep related data together to avoid cross-shard joins.



MEMORY HOOK

Sharding finds the data. Load balancing finds a healthy copy.

